OSI: It standardizes the function of a telecommunication or a computing system in 7 abstract layers.

Physical: physical connection between devices

Data-link: data transfer from a device to another within the same network (e.g. Switch).
Use MAC-address

Network: forwarding data from a network to another (IP-addressing)

Transport: correct forwarding of data from a program on your PC to a program on another PC. Data segments. Puts them back later.
TCP & UDP.

Session: responsible for establishing, managing, and terminating a communication session between two applications.

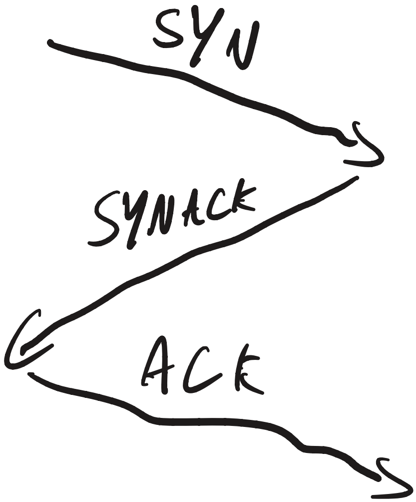
Presentation: ensures data is in the correct format and can be understand by both systems.

Application: interface between user's application and the network.

TCP vs UDP

→ TCP

Transmission Control Protocol. Slow but reliable. Use a three-way handshake.



e.g. HTTP, FTP, SSH

If a packet gets lost. TCP sends it back

→ UDP

Faster but not reliable. E.g. Video-meeting can accept latency or some packet-loss.

IP - addressing

→ /24 → .0 : Network address

→ .255 : Broadcast address

→ rest : usable hosts.

DNS (Domain Name System)

→ Converts an human-readable name into a IP-address

DNS Query to DNS Resolver → Checks cache

→ Empty → Root server → Where are all .com?

→ Goes to .com server and asks for the site → Sends IP-add back → 3-way handshake with the website

DHCP (Dynamic Host Configuration Protocol)

→ DHCP discover to see if there is a DHCP Server → DHCP offer (gives an IP)

→ DHCP Request to say you want that IP

→ DHCP Ack confirms and sends IP Default-gateway and DNS.

HTTP/S (Hypertext Transfer Protocol/Secure)

→ DNS Request → 3 way handshake

→ HTTP Request → HTTP Response

Secure adds a TLS encryption step.

Firewall & ACL : Stateless / Statful

→ Firewall is device that does what ACL wants

→ Stateless Packet Filtering : Looks at a packet simple information. Doesn't remember, every action is independent / no history

→ Stateful Packet Filtering : Has history, remembers everything e.g. if a person enters and leaves he remembers it's the same person that entered and left.

VPN: IPsec (L3) vs SSL-VPN (L7)

VPN creates a a private tunnel between 2 endpoints so that no one in the middle can read it.

IPsec: site-to-site VPN. Connects two office networks permanently.

SSL: Application layer. Uses TLS. For remote work. Connects one person to a network temporarily